VANISHING OF THE COTANGENT COMPLEX FROM TOR VANISHING AND VANISHING OVER THE TENSOR SQUARE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement carries two hypotheses (Tor vanishing and vanishing over the tensor square); the proof is a short computation using one base change and one distinguished triangle.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- Flat base change for the cotangent complex. Lemma lemma-flat-base-change-cotangent-complex (1201-1209), used at proof lines 1811-1813. Identifies the base change of L_{B/A} along B -> B tensor_A B.
- The distinguished triangle for the composition B -> B tensor_A B -> B. equation-triangle, used at proof lines 1814-1822. Is the triangle whose terms are shown to vanish.
- The vanishing of L_{B/B}. Lemma lemma-cotangent-complex-polynomial-algebra, used at proof lines 1820-1822. Gives one of the two vanishing inputs.
- The Tor vanishing hypothesis and the assumed vanishing over the tensor square. used at proof lines 1806-1807 and 1820-1822. Give the other vanishing input and make the base change computation meaningful.

Source reference. Original equation 08QS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11070–11070. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08QH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11060–11060. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08QQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11068–11068. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Let $A \to B$ be a ring map such that $Tor_i^A(B, B) = 0$ for $i > 0$ and such that $L_{B/B \otimes_A B} = 0$. Then $L_{B/A} = 0$ in $D(B)$.*

Proof. By Lemma 08QQ we see that $L_{B/A} \otimes^{\mathbf{L}}_B (B \otimes_A B) = L_{B \otimes_A B/B}$. Now we use the distinguished triangle (08QS)

$$L_{B \otimes_A B/B} \otimes^{\mathbf{L}}_{(B \otimes_A B)} B \to L_{B/B} \to L_{B/B \otimes_A B} \to L_{B \otimes_A B/B} \otimes^{\mathbf{L}}_{(B \otimes_A B)} B[1]$$

associated to the ring maps $B \to B \otimes_A B \to B$ and the vanishing of $L_{B/B}$ (Lemma 08QH) and $L_{B/B \otimes_A B}$ (assumed) to see that

$$0 = L_{B \otimes_A B/B} \otimes^{\mathbf{L}}_{(B \otimes_A B)} B = L_{B/A} \otimes^{\mathbf{L}}_B (B \otimes_A B) \otimes^{\mathbf{L}}_{(B \otimes_A B)} B = L_{B/A}$$

as desired. □